
Problem Set 8

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April 16, 2025

PY 355 Section A1

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Problem 1

1.

$$\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & b+a \\ 0 & 1 \end{pmatrix}.$$

2.

$$\det \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} = 1.$$

$$\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}^{-1} = \frac{1}{1} \begin{pmatrix} 1 & -a \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -a \\ 0 & 1 \end{pmatrix}.$$

3.

$$[M_a, M_b] = M_a M_b - M_b M_a = \begin{pmatrix} 1 & a+b \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & b+a \\ 0 & 1 \end{pmatrix} = 0.$$

4. Inducting on part 1 gives

$$\prod_{n=0}^N M_{2^n} = \begin{pmatrix} 1 & \sum_{n=0}^N 2^n \\ 0 & 1 \end{pmatrix}.$$

The series is geometric, and finite geometric series satisfy

$$\sum_{n=1}^N ar^n = a \left(\frac{1-r^{N+1}}{1-r} \right).$$

It follows that

$$\sum_{n=0}^N 2^n = \sum_{n=1}^{N+1} 2^{n-1} = \frac{1}{2} \sum_{n=1}^{N+1} 2^n = \frac{1}{2} \cdot 2 \left(\frac{1-2^{N+1}}{1-2} \right) = 2^{N+1} - 1.$$

Therefore,

$$\prod_{n=0}^N M_{2^n} = \begin{pmatrix} 1 & 2^{N+1} - 1 \\ 0 & 1 \end{pmatrix}.$$

Problem 2

1. I have absolutely no idea what the ϵ is supposed to be.

ok a quick google search shows that ϵ_{ijk} is the Levi-Civita symbol, which can be intuitively understood to behave like a wedge product or an alternating tensor, in that $\epsilon_{ijk} = 1$ if $ijk = 123$, and is alternating in that each transposition induces a factor of -1 , and this

anticommutativity necessarily implies $\epsilon_{ijk} = 0$ if $i = j$, $j = k$, or $k = i$. Due to this, the identity claims that $\sigma_i \sigma_i = 0 + \delta_{ii} 1 = 1$, so we verify this:

$$\begin{aligned} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \\ \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} &= \begin{pmatrix} -i^2 & 0 \\ 0 & -i^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \\ \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

This verifies the identity for $i = j$. Moreover, we now have

$$\sigma_i \sigma_j + \sigma_j \sigma_i = 2\delta_{ij} 1.$$

Note that

$$\begin{aligned} \sigma_i \sigma_j &= \sigma_i \sigma_j + \frac{1}{2} \sigma_j \sigma_i - \frac{1}{2} \sigma_j \sigma_i \\ &= \frac{1}{2} (\sigma_i \sigma_j - \sigma_j \sigma_i) + \frac{1}{2} (\sigma_i \sigma_j + \sigma_j \sigma_i) \\ &= \frac{1}{2} [\sigma_i, \sigma_j] + \frac{1}{2} 2\delta_{ij} 1 \\ &= \frac{1}{2} [\sigma_i, \sigma_j] + \delta_{ij} 1. \end{aligned}$$

It thus suffices to show $1/2 [\sigma_i, \sigma_j] = i\epsilon_{ijk} \sigma_k$. If we can show that $[\sigma_1, \sigma_2]$, $[\sigma_2, \sigma_3]$, and $[\sigma_3, \sigma_1]$ are equal to $2i\sigma_k$ for k the indicies not appearing, then this suffices as a proof, since anticommutivity of the commutator is trivially equivalent to the alternating nature of the Levi-Citiva symbol. These computations are simple and can be brute forced.

$$\begin{aligned} [\sigma_1, \sigma_2] &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} - \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} = 2i\sigma_3. \\ [\sigma_2, \sigma_3] &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} = 2i\sigma_1. \\ [\sigma_3, \sigma_1] &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix} = 2i\sigma_2. \end{aligned}$$

Therefore,

$$\sigma_i \sigma_j = i\epsilon_{ijk} \sigma_k + \delta_{ij} 1.$$

2. We have

$$(\sigma \cdot p)^2 = (\sigma_i p_i)^2 = \sigma_i p_i \sigma_j p_j = p_i p_j \sigma_i \sigma_j.$$

Using our previous result, we have

$$p_i p_j (i\epsilon_{ijk} \sigma_k + \delta_{ij} 1).$$

Evaluating the term on the right, we have

$$p_i p_j \delta_{ij} 1 = p_i p_i 1 = p_i^2 1 = |p|^2 1.$$

Evaluating the term on the left, we have

$$ip_i p_j \epsilon_{ijk} \sigma_k.$$

Note that for each term (i, j) , there is a corresponding term (j, i) , and the anticommutativity induced by the alternating Levi-Civita term causes these off-diagonal terms to vanish. Similarly, by anticommutativity once again, the diagonal terms also vanish, and thus

$$ip_i p_j \epsilon_{ijk} \sigma_k = 0.$$

Therefore,

$$(\sigma \cdot p) = |p|^2 1.$$

Problem 3

Let $A^2 = 0$. Then

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{pmatrix} = 0.$$

If $b = c = 0$, then clearly $A = 0$, so we can without loss of generality ignore this case. We have

$$b(a + d) = c(a + d) = 0.$$

Since we take at least one of $\{b, c\}$ to be nonzero, we must have $a + d = 0$. The matrix becomes

$$A = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}, \quad A^2 = \begin{pmatrix} a^2 + bc & ab - ab \\ ac - ac & a^2 + bc \end{pmatrix} = \begin{pmatrix} a^2 + bc & 0 \\ 0 & a^2 + bc \end{pmatrix}.$$

Now we re-parameterize with x and y . Choose $y \in \mathbb{C}$ such that $b = y^2$. Define $x := a/y$, to enforce that $a = xy$. We then have

$$(xy)^2 + y^2 c = y^2 (x^2 + c) = 0.$$

It follows that $c = -x^2$, since $b \neq 0$ by assumption. Therefore, our matrix becomes

$$A = \begin{pmatrix} xy & y^2 \\ -x^2 & -xy \end{pmatrix}.$$

This differs from the matrix in the problem only in notation.

Problem 4

1.

$$\det \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} = \det \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -1.$$

2.

$$\det \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix} = 1^2 - 6 = -5.$$

3.

$$\det \begin{pmatrix} 0 & \sqrt{2} & 0 & 0 \\ \sqrt{2} & 0 & 3 & 0 \\ 0 & 3 & 0 & \sqrt{2} \\ 0 & 0 & \sqrt{2} & 0 \end{pmatrix} = -\det \begin{pmatrix} \sqrt{2} & 3 & 0 \\ 0 & 0 & \sqrt{2} \\ 0 & \sqrt{2} & 0 \end{pmatrix} = \sqrt{2} \det \begin{pmatrix} 0 & \sqrt{2} \\ \sqrt{2} & 0 \end{pmatrix} = 2\sqrt{2}.$$

Problem 5

Determinants of triangular matrices are simply the product of the diagonal entries, and $\det AB = \det A \det B$. We have

$$\det A = (14 \cdot 3)(-8) = -42 \cdot 8 = -336.$$